Variational Quantum Eigensolver via Pauli Propagation

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Abstract

We implement a Variational Quantum Eigensolver (VQE) using Pauli propagation as a classical simulation backend to study ground state energies of frustrated quantum spin systems. The method propagates the Hamiltonian backwards through a parameterized circuit as a truncated weighted sum of Pauli strings, enabling efficient simulation without exponential state-vector storage. We benchmark against exact DMRG energies on three systems of increasing difficulty: the 1D Transverse Field Ising model (n=50), the 1D J1-J2 Heisenberg chain (n=20), and the 2D J1-J2 Heisenberg model on 4×4 and 6×6 square lattices (PBC). We achieve sub-1% relative error on the 1D systems and **0.64% error** on the 2D 4×4 system at the Néel-to-VBS phase transition ($J_2/J_1 = 0.535$), demonstrating Pauli propagation as a viable approach for classically simulating VQE on frustrated 2D magnets.

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Introduction

Finding the ground state energy of a quantum many-body Hamiltonian is one of the central problems in condensed matter physics and quantum chemistry. Frustrated quantum magnets — systems where competing interactions prevent simultaneous energy minimisation — are of particular interest due to their rich phase diagrams and exotic phases. The 2D J_1-J_2 Heisenberg model on the square lattice is a canonical example, hosting Néel antiferromagnetic (AFM), valence bond solid (VBS), and stripe AFM phases as the ratio J_2/J_1 is varied.

The Variational Quantum Eigensolver (VQE) is a hybrid quantum-classical algorithm that minimizes the expectation value

$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle \tag{1}$$

over a parameterized ansatz circuit $U(\boldsymbol{\theta})$. The variational principle guarantees $E(\boldsymbol{\theta}) \geq E_0$, providing rigorous upper bounds on the ground state energy. However, classical simulation of VQE is hampered by the exponential cost of state-vector methods.

Pauli propagation offers an alternative: rather than tracking the quantum state, the observable H is propagated backwards through the circuit as a weighted sum of Pauli strings [1]. Truncating terms with small coefficients ($|c| < \delta_c$) controls the computational cost at the expense of approximation accuracy. This approach scales with the number of Pauli strings retained rather than exponentially with system size, making it tractable for system sizes beyond exact diagonalisation [1].

Methods

Pauli Propagation

The Hamiltonian H is written as a sum of Pauli strings with real coefficients: $H = \sum_k c_k P_k$ where $P_k \in \{I, X, Y, Z\}^{\otimes n}$. Each parameterized gate in the circuit is a Pauli rotation $R_P(\theta) = e^{-i\theta P/2}$. Propagating such a gate backwards through a Pauli string Q produces:

$$R_P(\theta)^\dagger \, Q \, R_P(\theta) = \cos(\theta) \, Q \; + \; i \sin(\theta) \, \frac{[P, Q]}{2} \tag{2}$$

Each gate thus splits one Pauli string into at most two, causing exponential growth. This is controlled by discarding any string whose coefficient satisfies $|c| < \delta_c$. After propagating through the full circuit, the expectation value is obtained by computing the overlap of the resulting Pauli sum with the reference state $|\text{ref}\rangle$. All computations use `PauliPropagation.jl` [1] with the `PauliSum` data structure.

Optimization: SPSA + Adam

Parameters $\boldsymbol{\theta}$ are optimised using Simultaneous Perturbation Stochastic Approximation (SPSA) combined with Adam momentum [2]. SPSA estimates the gradient using only two function evaluations: a random perturbation vector $\boldsymbol{\Delta}$ is drawn, and the gradient is approximated as

$$\mathbf{g} \approx \frac{E(\boldsymbol{\theta} + \delta \boldsymbol{\Delta}) - E(\boldsymbol{\theta} - \delta \boldsymbol{\Delta})}{2\delta} \cdot \boldsymbol{\Delta} \tag{3}$$

Adam then applies adaptive per-parameter learning rates with momentum. Mid-run checkpoints save the best parameters every 100 iterations, allowing recovery if the job hits the wall-clock limit.

Hamiltonians

1D Transverse Field Ising (TFI)

$$H = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i \tag{4}$$

The dimensionless parameter $g_x = h/J$ controls the ratio of transverse field to coupling. A quantum phase transition occurs at $g_x = 1.0$, separating a ferromagnetically ordered phase ($g_x < 1$) from a disordered paramagnetic phase ($g_x > 1$). We study $n = 50$ sites with open boundary conditions (OBC) and $\delta_c = 10^{-5}$.

1D and 2D J_1-J_2 Heisenberg

$$H = J_1 \sum_{\langle i,j \rangle} \vec{\sigma}_i \cdot \vec{\sigma}_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} \vec{\sigma}_i \cdot \vec{\sigma}_j, \qquad \vec{\sigma}_i \cdot \vec{\sigma}_j = X_i X_j + Y_i Y_j + Z_i Z_j \tag{5}$$

J_1 and J_2 couple nearest-neighbor (NN) and next-nearest-neighbor (NNN) pairs respectively, both antiferromagnetic. The ratio J_2/J_1 controls the degree of magnetic frustration.

Circuit Ansätze

TFI – HVA-inspired ansatz with symmetry breaking. Each layer applies $R_Z \cdot R_X$ single-qubit rotations followed by R_{ZZ} on all NN bonds. The ansatz is inspired by the HVA, with R_X and R_{ZZ} matching the transverse field and coupling terms of the Hamiltonian respectively. The additional R_Z rotation is not part of the standard HVA but is included to break the $\mathbb{Z}_2$ symmetry of the Hamiltonian, allowing the circuit to reach the symmetry-broken ground state.

1D J_1-J_2. Each layer applies $R_Z \cdot R_X$ single-qubit rotations to all sites, followed by $R_{XX} \cdot R_{YY} \cdot R_{ZZ}$ on all NN bonds then all NNN bonds. Gate types mirror the Heisenberg interaction structure (HVA). R_Z and R_X together generate all single-qubit rotations.

2D J_1-J_2. Each layer applies $R_{XX} \cdot R_{YY} \cdot R_{ZZ}$ on NN bonds then NNN bonds, with no single-qubit layers, preserving SU(2) symmetry. Six shared parameters per layer ($\alpha_{\text{NN}}, \beta_{\text{NN}}, \gamma_{\text{NN}}, \alpha_{\text{NNN}}, \beta_{\text{NNN}}, \gamma_{\text{NNN}}$) — all bonds of the same type share one angle. Sites are labeled in snake ordering across the 2D lattice.

Reference States

- **TFI:** $|+\rangle^{\otimes n}$, the ground state of the pure transverse field limit.
- **1D J_1-J_2:** Majumdar-Ghosh (MG) dimer state — a product of singlet pairs $|\text{MG}\rangle = |s\rangle_{12} \otimes |s\rangle_{34} \otimes \cdots$, where $|s\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$. This is the *exact* ground state at $J_2/J_1 = 0.5$, explaining near-zero VQE error there.
- **2D J_1-J_2:** Néel state (checkerboard $\uparrow\downarrow$ pattern) for $J_2/J_1 \leq 0.535$; stripe state (alternating rows of $\uparrow$ and $\downarrow$ spins) for $J_2/J_1 > 0.535$. A columnar dimer reference was also tested but discarded — it produced 33–41% error, indicating the circuit could not deform it into the correct ground state.

DMRG Benchmark

All VQE energies are compared to DMRG ground state energies computed with `ITensors.jl`. For the 4×4 PBC system, max bond dimension 256 was used. For the 6×6 PBC system, max

bond dimension 550 was used, computed on GPU by the supervisor. The metric is relative energy error:

$$\frac{\Delta E}{|E_{\mathrm{DMRG}}|} = \frac{|E_{\mathrm{VQE}} - E_{\mathrm{DMRG}}|}{|E_{\mathrm{DMRG}}|} \tag{6}$$

Results

1D Transverse Field Ising ($n = 50$, OBC)

Figure 1 shows the relative energy error and Operator Stabilizer Entropy (OSE S_2) [3, 4] as a function of $g_x = h/J$ for $L = 5$ and $L = 10$ circuit layers with $\delta_c = 10^{-5}$. Both quantities peak near the critical point $g_x = 1.0$, where the ground state has maximum long-range entanglement and the Pauli sum is most complex. Away from the critical point, errors fall below 0.5% for $L = 10$. $L = 10$ consistently outperforms $L = 5$, as expected from greater circuit expressibility.

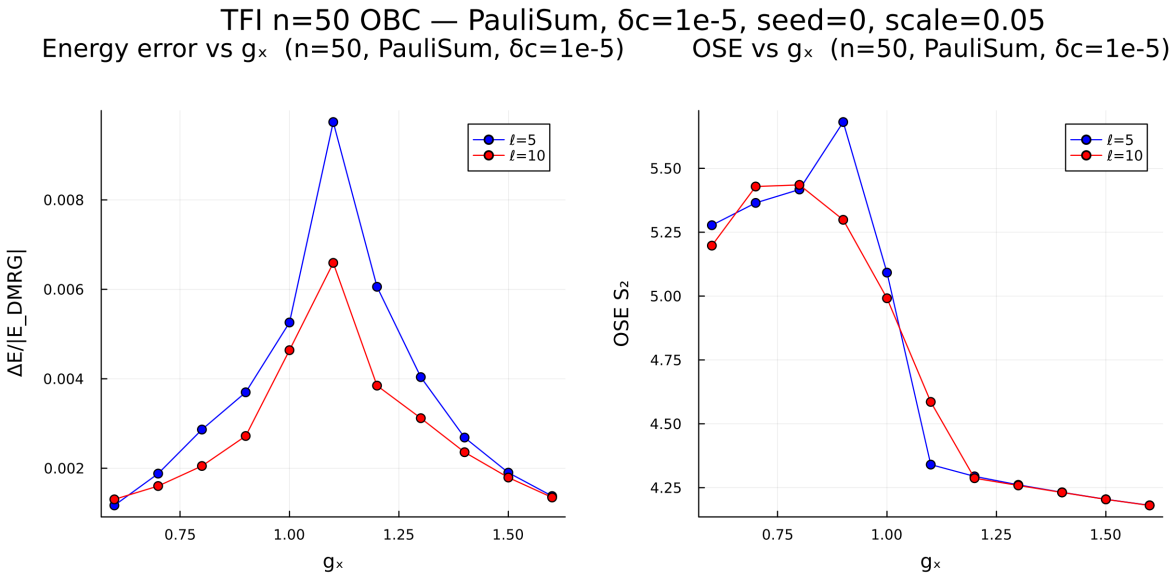

Figure 1: 1D TFI ($n = 50$, OBC, $\delta_c = 10^{-5}$). Left: relative energy error $\Delta E/|E_{\mathrm{DMRG}}|$ vs g_x. Right: OSE S_2 vs g_x. Both $L = 5$ and $L = 10$ shown.

1D J_1-J_2 Heisenberg ($n = 20$, OBC, MG reference)

Figure 2 shows results for the 1D frustrated Heisenberg chain with the MG dimer reference state. Near-zero error is achieved at $J_2/J_1 = 0.5$ — the Majumdar-Ghosh point where the MG state is the exact ground state and the circuit requires essentially no correction. As J_2/J_1 moves away from 0.5 in either direction, the true ground state changes character and the circuit must work harder to deform the MG reference into the correct ground state, causing error to rise. This is a natural consequence of the reference state choice: the further we are from the MG point, the more the circuit layers must compensate, and with only $L = 5$ or $L = 10$ layers this becomes increasingly difficult. The OSE should be interpreted with care here — further from $J_2/J_1 = 0.5$ the larger VQE errors mean the propagated observable is itself less accurate, so the OSE values in those regions reflect a less reliable approximation.

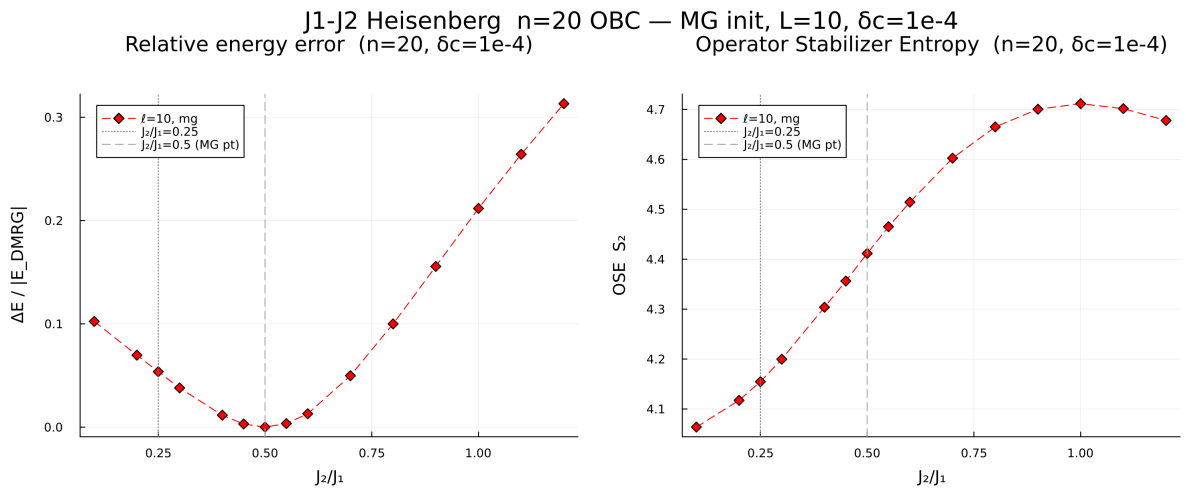

Figure 2: 1D J1-J2 Heisenberg ($n = 20$, OBC, $\delta_c = 10^{-4}$, MG dimer reference). Left: relative energy error. Right: OSE S_2. $L = 5$ and $L = 10$ shown.

2D J_1-J_2 Heisenberg — 4×4 PBC (16 qubits)

The 2D square lattice J_1-J_2 model hosts three phases [5]: Néel AFM ($J_2/J_1 < 0.535$), VBS ($0.535 < J_2/J_1 < 0.610$, nature debated [5]), and stripe AFM ($J_2/J_1 > 0.610$). Figure 3 shows exact VQE energies (solid lines) and partial $\tilde{f}$ estimates (dotted lines) for the 4×4 system with $L = 5$ and $\delta_c = 3 \times 10^{-4}$.

Best result: 0.64% relative error at $J_2/J_1 = 0.535$ with Néel initialisation — the Néel-to-VBS phase transition. Error is 6.6% at $J_2/J_1 = 0.3$ and 7.8% at $J_2/J_1 = 0.1$. The columnar dimer reference failed completely (33–41% error). Only 5 of 10 tasks completed within the 100-hour wall-clock limit.

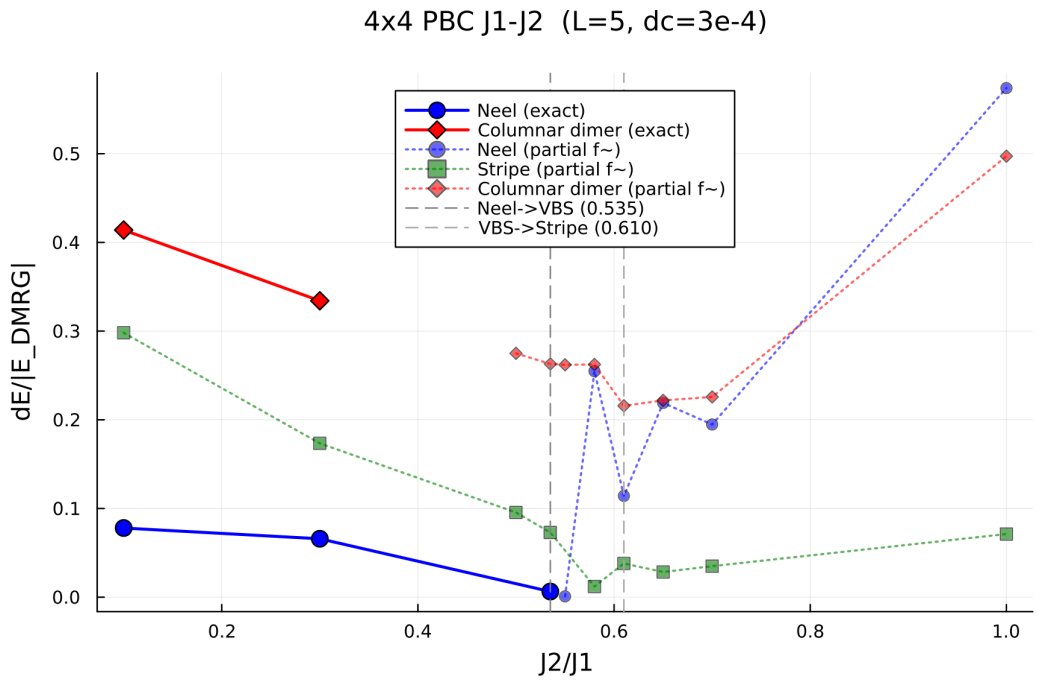

Figure 3: 2D J1-J2 4×4 PBC (16 qubits, $L = 5$, $\delta_c = 3 \times 10^{-4}$). Relative energy error $\Delta E/|E_{\mathrm{DMRG}}|$ vs J_2/J_1. Solid: exact evaluated energies. Dotted: partial $\tilde{f}$ estimates. Phase boundaries at $J_2/J_1 = 0.535$ and 0.610 marked.

2D J_1-J_2 Heisenberg — 6×6 PBC (36 qubits)

The 6×6 system presents a significant scaling challenge: 36 qubits, 72 NN and 72 NNN bonds, with Pauli sums reaching $\sim 10^8$ terms even at $\delta_c = 3 \times 10^{-4}$. The optimization ran for 100 hours on the ETH Zürich Euler cluster, with tasks reaching 1000–2700 of 3000 target iterations before the wall-clock limit. Saved parameters were then re-evaluated at $\delta_c = 10^{-5}$, providing proper variational upper bounds.

Figure 4 shows both the coarse $\delta_c = 3 \times 10^{-4}$ estimates (dotted) and the re-evaluated $\delta_c = 10^{-5}$ energies (solid). The dotted lines dip below DMRG for several points — this is a consequence of truncation altering the energy landscape rather than a true variational bound [6]. The solid $\delta_c = 10^{-5}$ lines all sit above DMRG as expected, with errors ranging from 6–17%, reflecting the partially converged optimization parameters. The frustrated region near the VBS phase ($J_2/J_1 \approx 0.535$–0.610) shows the largest errors, consistent with the increased difficulty of simulating this region.

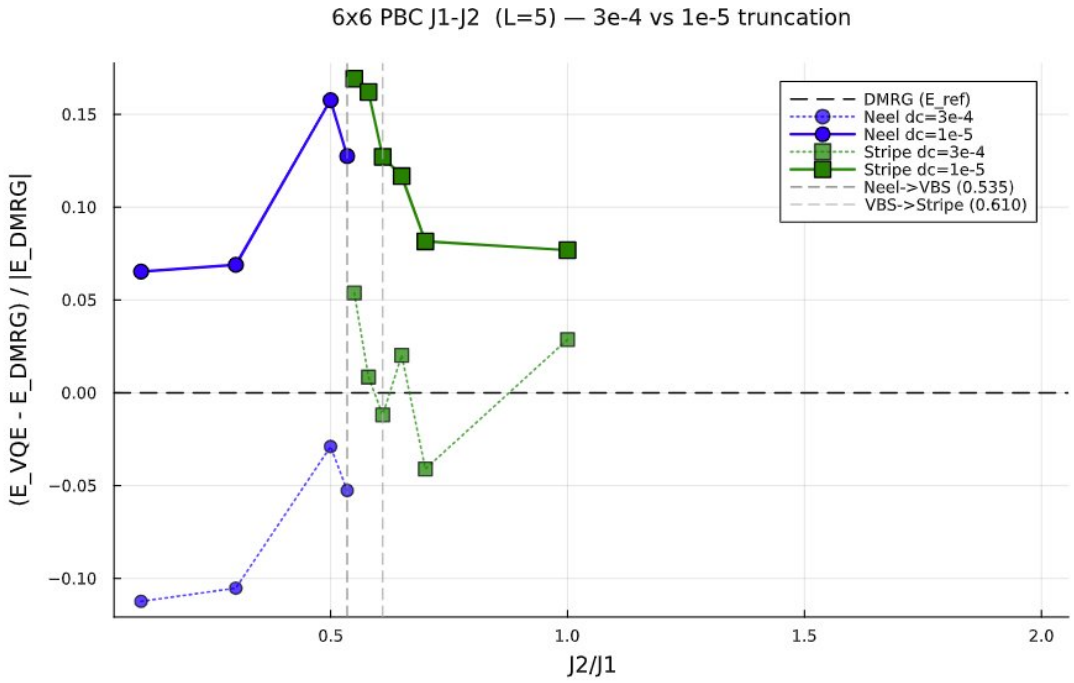

Figure 4: 2D J1-J2 6×6 PBC (36 qubits, $L = 5$). Signed relative error $(E_{\mathrm{VQE}}-E_{\mathrm{DMRG}})/|E_{\mathrm{DMRG}}|$ vs J_2/J_1. Dotted: coarse $\delta_c = 3 \times 10^{-4}$ SPSA estimates (points below zero are unphysical [6]). Solid: re-evaluated at $\delta_c = 10^{-5}$ using saved parameters — all above DMRG as expected. Phase boundaries at $J_2/J_1 = 0.535$ and 0.610 marked.

Discussion

Reference state choice is critical. The MG dimer state's exact optimality at $J_2/J_1 = 0.5$ produces near-zero error with minimal circuit depth, while the columnar dimer reference completely fails for 2D. Physically motivated reference states that capture the dominant order in each phase are essential for efficient VQE convergence.

Truncation and accuracy trade-off. The truncation parameter δ_c presents a fundamental trade-off: finer truncation retains more Pauli strings and yields more accurate energy estimates, but at the cost of significantly greater memory and runtime. Coarser truncation allows faster optimization over more iterations but produces a modified energy landscape that, as shown by [6], can cause the estimated energy to fall below the true ground state — explaining the unphysical points seen in the 6×6 results. Choosing the right δ_c therefore requires balancing simulation fidelity against computational feasibility, and is a key practical consideration when scaling to larger systems.

Scaling to 6×6. The 6×6 system highlights the memory challenge of Pauli propagation at scale. Pauli sums reach 10^8–10^9 terms at fine truncation for 36 qubits, exceeding 125 GB even with `VectorPauliSum` distributing across 10 threads. Future work could explore adaptive truncation schemes, Pauli sum compression, or running optimization directly at finer cutoff on higher-memory hardware.

OSE as a difficulty metric. The Operator Stabilizer Entropy S_2 [3, 4, 7] consistently tracks simulation difficulty, peaking at phase transitions and in frustrated regions. This suggests OSE could serve as an *a priori* indicator of where Pauli propagation will struggle and where circuit depth needs to be increased.

Conclusion

We have implemented VQE via Pauli propagation for three quantum spin systems of increasing complexity. The method achieves **0.64% relative error** on the 2D 4×4 J1-J2 Heisenberg model at the Néel-to-VBS phase transition, validating Pauli propagation as a tractable simulation approach for frustrated magnets. The 6×6 results show 6–17% errors after re-evaluation at $\delta_c = 10^{-5}$, reflecting partially converged parameters from the 100-hour optimization run.

Key findings:

1. Physically motivated reference states and ansatze are essential. For the 1D J_1-J_2 model, the MG dimer state is the exact ground state at $J_2/J_1 = 0.5$ and gives near-zero error there, with error growing as the system moves away from this point. For the 2D model, the Neel and Stripe states correctly capture the dominant magnetic order in their respective phases. In contrast, the columnar dimer reference completely fails in 2D (33–41% error), demonstrating that a poor reference state cannot be rescued by the circuit alone.
2. SPSA+Adam provides robust gradient-free optimization compatible with truncated Pauli sums, with mid-run checkpointing enabling recovery at wall-clock limits.
3. The OSE S_2 tracks simulation difficulty and may guide future resource allocation across phase diagrams.

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